Effect of the Nurse Work Environment on Older Hispanic Surgical Patient Readmissions

Marguerite Daus ▼ Matthew D. McHugh ▼ Ann Kutney-Lee ▼ Margo J. Brooks Carthon

Background: Readmissions following hospitalization for common surgical procedures are prevalent among older adults and are disproportionally experienced by Hispanic patients. One potential explanation for these disparities is that Hispanic patients may receive care in hospitals with lower-quality nursing care.

Objectives: The objective of this study was to evaluate the relationship between the hospital-level work environment of nurses and hospital readmissions among older Hispanic patients.

Methods: Using linked data sources from 2014 to 2016, we conducted a cross-sectional analysis of 522 hospitals and 732,035 general, orthopedic, and vascular surgical patients (80,978 Hispanic patients and 651,057 non-Hispanic White patients) in four states. Multivariable logistic regression models were employed to determine the relationship between the work environment and older Hispanic patient readmissions at multiple time periods (7, 30, and 90 days).

Results: In final adjusted models that included an interaction between work environment and ethnicity, an increase in the quality of the work environment resulted in a decrease in the odds of readmission that was greater for older Hispanic surgical patients at all time periods. Specifically, an increase in three of the five work environment subscales (Nurse Participation in Hospital Affairs, Nursing Foundations for Quality of Care, and Staffing and Resource Adequacy) was associated with a reduction in the odds of readmission that was greater for Hispanic patients than their non-Hispanic White counterparts.

Discussion: System-level investments in the work environment may reduce Hispanic patient readmission disparities. This study's findings may be used to inform the development of targeted interventions to prevent hospital readmissions for Hispanic patients.

Key Words: ethnic groups • nursing • occupational health • patient readmission • working conditions

Nursing Research, January/February 2024, Vol 73, No 1, E1–E10

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Hispanic people are the fastest-growing population in the United States (Passel et al., 2022). In 2020, almost 19% of the U.S. population identified as Hispanic; this number is projected to increase to 28% by 2060 (U.S. Census Bureau, 2018). As the Hispanic population has grown, evidence of health disparities between Hispanic and non-Hispanic White patients has surfaced, including postsurgical hospital readmissions. Hispanic patients are 15%–48% more likely to be readmitted following common surgical procedures (Aseltine et al., 2019; Oronce et al., 2015). These disparities may be attributed to the quality of hospitals where Hispanic patients receive care or decreased access to necessary services during hospitalization (Barnett & Vornovitsky, 2016; Jha et al., 2011). One characteristic of poorer quality may be insufficient nursing resources, such as the work environment, which can cause adverse outcomes when clinical care needs go unmet. Nurses are the largest contributors to direct patient care in hospitals, yet their influence on Hispanic surgical patients' outcomes is unknown.

Hispanic patients undergoing surgery are more likely to be readmitted for certain procedures (Aseltine et al., 2019; Oronce et al., 2015; Siracuse et al., 2017). These health disparities are attributed to complex, intersecting factors such as the social determinants of health and racism or unjust treatment and oppression of people of color, including Hispanic and Latino people (Braveman et al., 2022; Braveman & Gottlieb, 2014). In this study, we conceptualized ethnicity as a social construct that can be examined as a proxy for the effects of systemic racism and discrimination. These inequities may contribute to heightened risk among Hispanic surgical patients and elevated postsurgical complication risk. One result of racism is that Hispanic patients are more often treated in low-quality, high-cost hospitals

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DOI: 10.1097/NNR.0000000000000698

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with poor patient outcomes (Jha et al., 2011). These are mainly small, public hospitals in the South with poorer nurse staffing levels. These findings may indicate decreased access to adequate nursing resources, such as the work environment, that reduce poor postsurgical outcomes (Brooks Carthon et al., 2012; Lasater & McHugh, 2016a; Ma et al., 2015). Because healthcare systems determine what resources are available to patients, it is essential to examine the characteristics of hospitals as they relate to hospital nursing.

Nursing care is fundamental to preserving functional status, preventing complications, and enhancing surgical patient recovery. Additional nursing considerations for Hispanic surgical patients include assessing cultural preferences, language-concordant care, family involvement in decisions, and time to build trust (Martinez Tyson et al., 2017). Poorer outcomes, including higher rates of hospital readmission, result when these care needs are unmet by nurses (Brooks Carthon et al., 2013; Lasater & McHugh, 2016a; Ma et al., 2015). Nurses' abilities to provide tailored care are largely associated with the support they receive in their clinical settings, which the work environment can measure. Improvements in the work environment may reduce the likelihood of surgical patient rehospitalization (Ma et al., 2015; McHugh & Ma, 2013). Populations experiencing disparities may be particularly sensitive to variations in nursing care quality as older Black patients treated in hospitals with optimal nurse staffing and work environments had better outcomes, including readmissions (Brooks Carthon et al., 2013, 2021; Lasater & McHugh, 2016b). We hypothesize that optimizing nursing resources in settings where Hispanic patients receive care may help reduce disparate outcomes, but this relationship is poorly understood.

The aims of the study were (a) to determine if there were differences in the odds of hospital readmissions at 7, 30, and 90 days between Hispanic surgical patients and their non-Hispanic White counterparts and if those differences were conditional on the work environment and (b) to explore whether differences in the odds of hospital readmission between Hispanic and non-Hispanic White surgical patients were conditional on the individual work environment subscales. We contribute to scientific knowledge by examining health disparities and the work environment—a modifiable system factor—to inform intervention development to reduce preventable Hispanic patient readmissions.

METHODS

Design

This cross-sectional secondary data analysis comprised 522 hospitals in four states (California, Florida, New Jersey, and Pennsylvania) and included 732,035 surgical patients cared for in these hospitals from 2015 to 2016. Data were merged using a common hospital identifier to create a multilevel data set with patient state discharge records from four state registries (Pennsylvania Health Care Cost Containment Office, Florida's Agency for Health Care Administration, Office of Statewide Health Planning and Development in California, and The New Jersey Department of Health and Senior Services). Hospital characteristics were derived from the American Hospital Association (AHA) Annual Survey of Hospitals and the RN4CAST-US survey, collected from a random sample of registered nurses working in hospitals that asked about nurse work environment, nurse staffing, and nurse education.

Setting and Sample

Hospitals The hospital sample included adult, nonfederal, acute care hospitals represented in the nursing and patient data that had at least 10 RN4CAST-US nurse respondents, 50 surgical cases, and participated in the 2016 AHA Annual Survey (Aiken et al., 2008; McHugh & Ma, 2013). Following prior work, a minimum of 10 nurse respondents per hospital was required to produce reliable hospital-level estimates of nursing resources (Lasater et al., 2019; Ma et al., 2015). RN data (n = 26,050) came from the RN4CAST-US survey conducted at the University of Pennsylvania. A modified Dillman approach was used to recruit a random sample of 30% of nurses holding an active RN license and a mailing address in one of the four states between 2015 and 2016, with a total sample of approximately 230,000 nurses. The RN4CAST-US survey's response rate was 26%; additional information on survey methodology has been previously published (Lasater et al., 2019). The survey queried nurses regarding demographics, the work environment, and their employing hospital to link nurse data to patient and hospital data. On average, there were 39 nurse respondents (minimum–maximum = 10–204) per hospital. Additional hospital characteristics were collected from the 2016 AHA Annual Survey, including structural, organizational, and financial measures of hospitals that may be associated with patient outcomes.

Patients Patient characteristics and outcomes were derived using administrative claims data from four state registries, including clinical information such as diagnosis-related groups (DRGs), International Classification of Diseases (ICD) diagnosis codes, patient demographics, admission and discharge dates, and readmissions. Admissions from October 1, 2014, through September 30, 2014, included ICD-9 codes and ICD-10 codes for admissions on and after October 1, 2015. We limited the sample to (a) Hispanic and non-Hispanic White adults between the age of 65 and 99 years, (b) hospitalized between the fourth quarters of 2014 and 2016, (c) with a principal procedure type of general, orthopedic, or vascular surgery, and (d) with a length of stay of at least 1 day. These combined surgical groups were selected because they represent a large proportion of procedures conducted across most hospitals and are commonly examined together (Ma et al., 2015; Press et al., 2011). Patients were excluded if they died during the index hospitalization or were discharged against medical advice.

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Variables and Instruments

Outcome Variables Study outcomes included 7-, 30-, and 90-day hospital readmission following index admission for general, orthopedic, or vascular surgery. Readmission was defined as the first all-cause hospital admission following an index hospital discharge. The patient's first discharge for an eligible diagnosis in that period served as the index discharge based on the Centers for Medicare & Medicaid Services (CMS) Risk-Standardization Readmission Measures (Horwitz et al., 2011).

Explanatory Variables The explanatory variables were patient-level Hispanic ethnicity and hospital-level work environment. Hispanic ethnicity was derived from state discharge data, categorized by patients who were identified as Hispanic and those who were not. Although the definition of Hispanic is controversial, the most common way to define Hispanic patients in health services research is to use those who are identified as Hispanic in ethnicity data (Lopez et al., 2023).

Work environment was measured by the Practice Environment Scale–Nursing Work Index (PES-NWI), a National Quality Forum-endorsed and -validated 31-item scale included in the RN4CAST-US survey (Lake, 2002, 2007). The PES-NWI has five subscales: (a) Nurse Participation in Hospital Affairs; (b) Nursing Foundations for Quality of Care; (c) Nurse Manager Ability, Leadership, and Support of Nurses; (d) Staffing and Resource Adequacy; and (e) Collegial Nurse and Physician Relations. Nurse respondents rated the degree to which each item was present in their current jobs on a Likert scale (1 = *strongly disagree*, 4 = *strongly agree*). Responses were aggregated to create a mean response by hospital for each subscale. The global work environment measure represented the mean score of all hospital-level subscales. Each individual subscale of the PES-NWI was evaluated as an explanatory variable, using the mean score at the hospital level for each subscale. The PES-NWI and subscales were standardized, so the odds ratios (*OR*s) represented the effect of a 1 *SD* increase in the work environment on readmissions.

Hospital Covariates Structural hospital characteristics were derived from the 2016 AHA Annual Survey and included bed size, teaching status, technology status, state, and core-based statistical area. "Top 25% Medicaid" was a dichotomous variable representing hospitals in the top 25th percentile in their admissions for all Medicaid patients (Rodriguez et al., 2011). To isolate the effect of the work environment, we controlled for nursing resources associated empirically with readmissions and derived from the RN4CAST-US (Lasater & McHugh, 2016a; Ma et al., 2015). Nurse staffing was calculated from survey respondents' reports of the number of nurses providing direct patient care and the number of patients on the unit during their last shift (Aiken et al., 2011). Nurse education was aggregated at the hospital level as a continuous variable of the percentage of nurses with a baccalaureate degree in nursing (BSN) or higher (Aiken et al., 2011). The percentage of intensive care unit (ICU) nurse respondents was measured as the number of ICU nurse respondents in a hospital divided by the total number of direct care nurse respondents (Brooks Carthon et al., 2021).

Patient Risk Adjustment Patient covariates were derived from state discharge data and based on prior literature, including age, gender, race (for non-White Hispanic patients), socioeconomic status, procedure type, comorbidities, discharge disposition, and if the patient was transferred from an outside hospital (Ma et al., 2015; McHugh & Ma, 2013). Patient comorbidities were classified using Elixhauser's comorbidity index (Elixhauser et al., 1998). Two comorbidities, fluid and electrolyte disorders and coagulopathy, were excluded because of high rates of misclassification in concordance with previous literature (Quan et al., 2005). Procedure type was identified by 58 DRG codes included in our analyses that indicated what type of surgical procedure was performed. Socioeconomic status was measured by insurance status, where having a primary payor of Medicaid indicated low income (National Quality Forum, 2014).

Analysis

We used inferential statistics to examine the differences in hospital, nursing, and patient characteristics between older Hispanic and non-Hispanic White patients. We examined the distribution of patients by each hospital, nursing, and patient characteristic using two-sided *t*-tests and Pearson chi-square analyses to test for statistical significance. Sequential multivariate logistic regression models were fit to determine the association between readmissions, ethnicity, and the work environment using clustering to account for the similarities of patients treated within the same hospital. Different times to readmission (7, 30, and 90 days) were examined as the outcome for each model separately to determine whether the effects of nursing resources on readmissions varied at each period. Finally, interaction terms were included in the fully adjusted model that allowed us to isolate the differential effects of work environment between Hispanic and non-Hispanic White patients. To identify which subscale may be driving the relationship between ethnicity and the work environment, each of the five subscales was fit individually in separate models that followed the above process. We tested all statistics and regression models for significance with an alpha of .05 using STATA (Release 14; StataCorp, 2015).

Informed Consent and Ethical Approval

The RN4CAST-US survey was approved by the University of Pennsylvania Institutional Review Board, and consent was obtained from participating nurses. Patient informed consent was waived as patient data were deidentified and assigned a random ID number. This study was determined exempt by the University of Pennsylvania Institutional Review Board.

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RESULTS

Sample Characteristics

The final sample included 732,035 total patients, 80,978 of those identifying as Hispanic (11%). Demographic characteristics of our patient sample stratified by ethnicity are presented in Table 1. Hispanic patients (74.9%) were slightly younger than non-Hispanic White patients (75.6%). Both Hispanic (57%) and non-Hispanic White (56%) patients were more likely to be female. Hispanic patients identified mainly as White (63%) or other (34%). Less than 1% of Hispanic patients identified as Black, Asian, or Native American. Hispanic patients were significantly more likely to be insured solely by Medicaid (5% vs. <1%), discharged to home on self-care (48% vs. 40%), and undergo vascular (34% vs. 24%) or orthopedic (18% vs. 15%) surgery. Finally, Hispanic patients in our sample experienced a higher average number of comorbidities (M = 2.6, SD = 1.7) than their non-Hispanic White counterparts (M = 2.4, SD = 1.6). For all readmission outcome lengths (7, 30, and 90 days), Hispanic patients had higher readmission rates than non-Hispanic White surgical patients (e.g., at 30 days, 11% of Hispanic patients readmitted compared to 10% of White patients). All differences were significant at $p < .001$.

Hospital characteristics by ethnicity are included in Table 2. Hispanic patients were, on average, treated in hospitals with higher composite PES-NWI scores (2.82 vs. 2.77) and higher scores for every PES-NWI subscale compared to non-Hispanic White patients. Most Hispanic patients received care at larger, urban hospitals in California more likely to have high technology and teaching status. Hispanic patients were also located in hospitals that were more likely to be in the top quartile for percentage of Medicaid patients (62% vs. 12%). Lastly, Hispanic patients were treated in better-staffed hospitals (4.2 vs. 4.3) with higher average percentages of BSN prepared (62% vs. 58%) compared to non-Hispanic White patients. All differences were significant at $p < .001$.

Association of the Work Environment and Hospital Readmissions

We display the *OR* indicating the association of the work environment and ethnicity on the odds of 7-, 30-, and 90-day readmission

TABLE 1. Surgical Patient Characteristics by Hispanic Patient Ethnicity (N = 732,035)

Characteristic	All patients (N = 732,035)		Non-Hispanic patients (n = 651,057, 89%)		Hispanic patients (n = 80,978, 11%)		
Readmission outcomes							
	n	%	*n*	%	*n*	%	*p*
7-day readmission	28,561	3.9	24,992	3.8	3,569	4.4	<.001
30-day readmission	72,025	9.8	63,141	9.7	8,884	11.0	<.001
90-day readmission	122,029	16.7	107,492	16.5	14,537	18.0	<.001
Patient characteristics							
	Mean	*SD*	Mean	*SD*	Mean	*SD*	*p*
Age	75.5	7.7	75.6	7.7	74.9	7.4	<.001
Total comorbidities	2.4	1.6	2.4	1.6	2.6	1.7	<.001
	n	%	*n*	%	*n*	%	*p*
Female	411,054	56	364,675	56	46,379	57	<.001
Race							<.001
White	702,424	96	651,057	100.0	51,367	63	
Black	1,135	0.2		—	1,135	1.4	
Asian	617	0.1		—	617	0.8	
Native American	241	0.1		—	241	0.3	
Other	27,618	3.8		—	27,618	34	
Medicaid status	5,656	0.8	1,936	0.3	3,720	4.6	<.001
Transfer	17,096	2.3	15,174	2.3	1,922	2.4	.447
Discharge disposition							
Home	296,641	41	257,657	40	38,984	48	<.001
Home health	212,304	29	192,075	30	20,229	25	<.001
Short-term hospital	4,939	0.7	4,343	0.7	596	0.7	.024
SNF	216,300	30	195,436	30	20,864	26	<.001
Intermediate care	1,851	0.3	1,546	0.2	305	0.4	<.001
Surgery type							<.001
General	183,945	25	155,323	24	27,622	34	
Orthopedic	437,438	60	398,569	61	38,869	48	
Vascular	111,652	15	97,165	15	14,487	18	

Note. SNF = skilled nursing facility.

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TABLE 2. Hospital Characteristics by Hispanic Patient Ethnicity

	Hospitals		Non-Hispanic White patients		Hispanic patients		
	(*n* = 522)		(*n* = 651,057, 89%)		(*n* = 80,978, 11%)		
Characteristic	Mean	*SD*	Mean	*SD*	Mean	*SD*	*p*
NWI	2.74	0.2	2.77	0.2	2.82	0.2	<.001
NWI subscales							
Affairs	2.65	0.3	2.72	0.3	2.77	0.3	<.001
Foundations	3.04	0.2	3.09	0.2	3.14	0.2	<.001
MNALS	2.75	0.3	2.79	0.3	2.84	0.3	<.001
NP Relate	2.74	0.1	2.74	0.1	2.75	0.1	<.001
SRA	2.51	0.3	2.53	0.3	2.61	0.3	<.001
Nurse staffing	4.4	1.0	4.3	0.8	4.2	0.9	<.001
% BSN	56.1	15	57.8	14	61.5	13	<.001
% ICU nurses	16.2	8.3	16.4	7	18.0	9	<.001
	n	%	*n*	%	*n*	%	
NWI							<.001
Poor	131	25	129,253	19.8	10,564	13.0	
Mixed	261	50	307,828	47.3	41,509	52.3	
Good	130	25	213,976	32.9	28,905	35.7	
State							<.001
CA	206	39	210,547	32	46,038	57	
FL	149	29	237,303	36	29,107	36	
PA	110	21	133,416	20	1,068	1.3	
NJ	57	11	69,791	10	4,765	6	
Urban	488	93	634,716	98	80,629	99	<.001
High tech	279	53	449,755	69	57,885	72	<.001
Bed size							<.001
Small	39	7.5	15,373	2.4	478	0.6	
Medium	219	42	189,478	29	18,449	22	
Large	264	51	446,206	69	62,051	77	
Teaching status							<.001
None	223	42	280,170	43	30,318	37	
Minor	250	48	293,974	45	41,924	52	
Major	49	10	76,913	12	8,736	11	
Top 25% Medicaid	130	25	79,755	12	50,603	62	<.001

Note. BSN = bachelor's degree in nursing science; ICU = intensive care unit. Staffing is defined as the number of patients per nurse. NWI represents the work environment measured by the Practice Environment Scale of the Nursing Work Index (PES-NWI). Affairs represents Nurse Participation of Hospital Affairs Nurse Participation of Hospital Affairs; Foundations represents Nursing Foundations for Quality of Care; MNALS represents Nurse Participation of Nurse Manager Ability, Leadership, and Support of Nurses; NP Relate represents Collegial Nurse and Physician Relationships; SRA represents Staffing and Resource Adequacy, all of which are subscales of the PES-NWI. *n* = count.

with unadjusted models (Model 1), adjusting for patient and hospital characteristics (Model 2), and interaction models (Model 3) in Table 3. In unadjusted models for all time periods, Hispanic patients had higher odds of readmission than non-Hispanic White patients, and each 1 *SD* increase in work environment resulted in reduced readmission odds for all patients. Adjusting for patient and hospital characteristics, Hispanic patients—relative to non-Hispanic White patients—experienced 6%–11% higher odds of readmission at all time periods (7 days: $OR = 1.11$, $p < .001$; 30 days: $OR = 1.09$, $p < .001$; and 90 days: $OR = 1.06$, $p < .001$). For each 1 *SD* increase in the work environment, there was a 3% decrease in odds of readmission for all patients at 7 and 30 days ($OR = 0.97$, $p < .01$ for 7 and 30 days).

Interactions Between Hispanic Ethnicity and the Work Environment

At all time points, a 1 *SD* increase in hospital work environment lowered the odds of readmission more for Hispanic patients than non-Hispanic White patients. At 7 days, Hispanic patients experienced a 6% decrease ($0.96 \times 0.98 = 0.94$) in odds of readmission, whereas the odds of readmission for non-Hispanic White patients decreased by 2% ($OR = 0.98$, $p = .04$). At 30 days, the odds of readmission were lowered by 5% ($0.97 \times 0.98 = 0.95$) for Hispanic patients whereas their non-Hispanic White counterparts' odds decreased by 2% ($OR = 0.98$, $p = .02$). For 90-day readmissions, Hispanic patients experienced a 4% reduction ($0.97 \times 0.99 = 0.96$) in the odds of readmission compared to

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TABLE 3. Sequential Modeling Indicating the Effects of Ethnicity and Work Environment on Readmissions

	Unadjusted model		Adjusted models for patient and hospital characteristics			
	Model 1		Model 2		Model 3	
7-day readmission						
Hispanic ethnicity	1.15	[1.09, 1.21]***	1.11	[1.05, 1.18]***	1.13	[1.03, 1.16]***
Standardized NWI		—	0.97	[0.95, 0.99]**	0.98	[0.96, 0.99]*
Hispanic Ethnicity × Standardized NWI		—		—	0.96	[0.92, 0.99]*
30-day readmission						
Hispanic ethnicity	1.15	[1.10, 1.20]***	1.09	[1.05, 1.13]***	1.10	[1.06,1.14]***
Standardized NWI		—	0.97	[0.96, 0.99]**	0.98	[0.96, 0.99]*
Hispanic Ethnicity × Standardized NWI		—		—	0.97	[0.94, 0.99]*
90-day readmission						
Hispanic ethnicity	1.11	[1.06, 1.15]***	1.06	[1.03, 1.10]***	1.08	[1.04, 1.11]**
Standardized NWI		—	0.99	[0.97, 1.00]	0.99	[0.97, 1.00]
Hispanic Ethnicity × Standardized NWI		—		—	0.97	[0.94, 0.99]*

Note. Values represent odds ratios [95% confidence intervals]. The models above examine the likelihood of readmission adjusting for clusters of patient, hospital, and nursing characteristics. Model 1 is unadjusted and examines the univariate associations between Hispanic ethnicity and readmissions. Model 2 is multivariate and adjusted for nurse staffing, % BSN, hospital characteristics (bed size, teaching status, high tech, urban, state [PA, NJ, CA, FL], top 25% Medicaid), and patient characteristics (age, Hispanic patient race, gender, transfer status, discharge disposition, Elixhauser comorbid conditions, MS-DRGs, Medicaid status). Model 3 is adjusted for the same characteristics as Model 3 but includes an interaction term examining the differential effects of NWI between Hispanic and Non-Hispanic White patients on readmissions. NWI represents the work environment measured by the Practice Environment Scale of the Nursing Work Index (PES-NWI) and is standardized in all models.

*$p < .05$.

**$p < .01$.

***$p < .001$.

non-Hispanic White patients who experienced a 1% reduction with each 1 *SD* increase in the work environment (*OR* = 0.99, $p = .22$). The difference was more pronounced between the *OR* of readmission for Hispanic and non-Hispanic White patients in hospitals with lower PES-NWI scores. For ease of interpretation, predicted probabilities of 30-day readmission were calculated for Hispanic and non-Hispanic White patients in varying work environments. Figure 1 suggests that the probability of readmission for Hispanic patients is equal to that of non-Hispanic White patients when PES-NWI scores are highest (3.8–4), but Hispanic patients in hospitals with lower PES-NWI scores have higher probability of readmission than non-Hispanic White patients.

We present the results of analyses examining the relationships between the five individual PES-NWI subscales and ethnicity in Table 4. Three subscales were significantly related to the odds of Hispanic patient readmission: Hospital Affairs at 7 days (*OR* = 0.96, $p = .04$), Nursing Foundations at 30 and 90 days (30 days: *OR* = 0.96, $p = .01$; 90 days: *OR* = 0.96, $p = .01$), and Staffing and Resource Adequacy at 7 and 30 days (7 days: *OR* = 0.94, $p = .007$; 30 days: *OR* = 0.97, $p = .04$). The associated *OR*s indicate that differences in readmissions between Hispanic and non-Hispanic White patients are more pronounced for Hispanic patients in hospitals with poorer scores on those three subscales. We found no relationship between readmission and

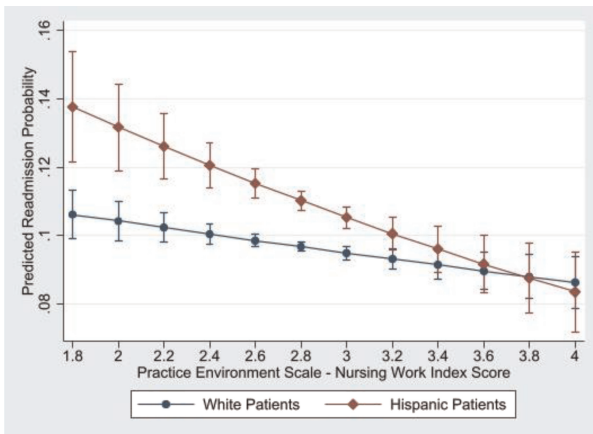

FIGURE 1. Predicted probability of readmission at 30 days for varying Practice Environment Scale–Nursing Work Index scores.

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the Nurse Manager and Leadership or the Nurse–Physician Relationship subscales.

DISCUSSION

In our evaluation of over 500 hospitals and 735,000 patients, older Hispanic surgical patients cared for in better work environments had lower odds of 7-, 30-, and 90-day readmissions. Our results are consistent with an expansive body of evidence linking better work environments to improved outcomes and recent findings suggesting that Black patients may be more sensitive to the quality of the work environment (Brooks Carthon et al., 2015; Lasater & McHugh, 2016b). Although other researchers have found that older Hispanic patients are often treated in poorer-quality hospitals, the Hispanic patients in our study were, on average, treated in hospitals with higher PES-NWI scores. However, 65% of Hispanic patients in our sample were treated in mixed or poor work environments, exposing them to higher risk for adverse outcomes and supporting our hypothesis that Hispanic patients may be more sensitive to variations in nursing resources.

Our findings provide translatable solutions for healthcare administrators to address hospital readmission disparities among older Hispanic surgical patients by focusing on improvements in the work environment. Hospitals with supportive work environments establish pathways for nurses to be patient advocates and listen to nurses when they believe a patient needs support (Lake, 2002). An organization that encourages autonomy empowers nurses to advocate for Hispanic patients and alert the appropriate providers when they detect physical or social needs that should be addressed (Gerber, 2018). These hospitals provide resources tailored to Hispanic patient needs so nurses can deliver the highest quality care.

Pursuing ANCC Magnet Recognition is one approach to improve the work environment. The ANCC Magnet Recognition Program designation is given to hospitals that align their nursing goals to improve the organization's patient outcomes (American Nurses Credentialing Center, n.d.). Researchers have found that Magnet hospitals report higher-quality work environments than non-Magnet hospitals and better patient outcomes, including readmissions (Kutney-Lee et al., 2015). Given the rigorous and costly pathway to Magnet, many hospitals may be unable to invest in the program. Our approach of examining individual PES-NWI subscales provides actionable insight for these hospitals on areas of low-cost investment to improve their work environments.

We found that three of the five NWI subscales had persistent, significant relationships with Hispanic patient readmissions. Nurse Participation in Hospital Affairs subscale measures nurses' value within the hospital context, as evidenced by their inclusion in policy development, shared governance, and as committee members. In these hospitals, nurses feel valued, listened to by administrators, and empowered to advocate for Hispanic patient needs. Nursing Foundations for Quality of Care considers if a nursing model of care is present that supports continuing education in the hospital. Hospital investment in their nurses' growth may contribute to improved patient outcomes through nurse-directed care, evidence-based practice, and innovation. These hospitals may provide cultural competence training and consider resources specific to Hispanic patients to ensure nurses deliver appropriate care. Finally, Staffing and Resource Adequacy describes the adequacy of nurse staffing and resources for nurses to safely deliver patient care. This gives more time for nurses to address culturally specific needs of Hispanic patients, such as language-concordant care, education on

TABLE 4. Conditional Effects of Individual NWI Subscales on Ethnicity Adjusted for Clusters of Patient, Hospital, and Nursing Characteristics

	7-day readmission		30-day readmission		90-day readmission	
Interaction	***OR***	**95% CI**	***OR***	**95% CI**	***OR***	**95% CI**
Hispanic × Affairs	0.96	[0.92, 0.99]*	0.97	[0.96, 1.00]	0.97	[0.94, 1.00]
Hispanic × Foundations	0.96	[0.92, 1.00]	0.96	[0.94, 0.99]*	0.96	[0.94, 0.99]*
Hispanic × MNALS	0.97	[0.93, 1.01]	0.97	[0.94, 1.00]	0.97	[0.94, 1.00]
Hispanic × N-P Relate	1.00	[0.94, 1.05]	0.98	[0.94, 1.03]	0.97	[0.94, 1.01]
Hispanic × SRA	0.94	[0.90, 0.98]**	0.97	[0.94, 0.99]*	0.98	[0.95, 1.01]

Note. The models above examine the likelihood of readmission adjusting for clusters of patient, hospital, and nursing characteristics. All models are multivariate and adjusted for staffing, % BSN, state (PA, NJ, CA, FL), hospital characteristics (bed size, teaching status, high tech, urban, state, top quartile % Medicaid), and patient characteristics (age, Hispanic patient race, gender, transfer status, discharge disposition, Elixhauser comorbid conditions, surgical DRGs, Medicaid status). They all include an interaction term examining the differential effects of the NWI subscale between Hispanic and Non-Hispanic White patients on readmissions. NWI represents the work environment measured by the Practice Environment Scale of the Nursing Work Index (PES-NWI) and is standardized. Affairs represents Nurse Participation of Hospital Affairs Nurse Participation of Hospital Affairs; Foundations represents Nursing Foundations for Quality of Care; MNALS represents Nurse Participation of Nurse Manager Ability, Leadership, and Support of Nurses; NP Relate represents Collegial Nurse and Physician Relationships; SRA represents Staffing and Resource Adequacy, all of which are subscales of the PES-NWI. *OR* = odds ratio; CI = confidence interval.

*$p < .05$.

**$p < .01$.

***$p < .001$.

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postdischarge care, and communication to build trust. Hospital administrators should target these components of the work environment to develop low-cost solutions to reduce older Hispanic surgical patient readmissions.

CMS Policy and Readmissions

Our findings that Hispanic patients are sensitive to fluctuations in the work environment anywhere from 7 to 90 days postdischarge have policy implications. Although CMS established 30-day readmissions as a hospital quality indicator tied to reimbursement, some researchers and administrators are proponents of shorter periods, such as 7 days, given the closer proximity to hospital care (Chin et al., 2016). The Bundled Payments for Care Improvement initiative trialed by CMS includes models where healthcare systems assume financial risk during the surgical inpatient stay and related services up to 90 days after hospital discharge (CMS, 2022). If the Bundled Payments for Care Improvement is widely adopted, hospitals may bear the cost of any readmission up to 90 days following discharge. Our study informs CMS policies and their consideration of Hispanic patients for 90 days as the episode of care window during which hospitals are liable for readmissions.

Combating Systematic Racism

At the crux of the disparities revealed in this study is systematic racism. Hence, system-level interventions are required that consider not only how to meet the current community needs but also how to transform healthcare delivery to ensure it is offered equitably. A hospital culture that centers on equitable outcomes and meaningfully tracks their progress is necessary to address health disparities. Although some hospitals invest in cultural competency training, translators, and multilingual patient instructions, this may be insufficient without adequate nursing resources and organizational support that enable nurses to utilize them. Our findings suggest that malleable, system-level investments in nursing resources could result in meaningful reductions in Hispanic patient readmissions, enhancing quality of life and decreasing hospital costs.

Limitations

Cross-sectional data limit our ability to determine causal relationships. However, we are confident in our results, given that other studies have shown similar relationships between the work environment and readmissions in minoritized populations (Brooks Carthon et al., 2015). This study includes only four states, which may limit generalizability; however, these states include over 20% of all acute care hospitals nationally and 35% of the U.S. Hispanic population. Our sample may not fully represent the four states' Hispanic population, as Hispanic patients may be undercounted or misrepresented because of the use of administrative billing data (Filice & Joynt, 2017). Analyses examining Hispanic patient outcomes according to race, country of origin, and indigenous identities were not possible.

In addition, our ability to detect and address health disparities is limited by data availability and aggregation. As the Hispanic population is diverse and heterogeneous, the need for disaggregated race and ethnicity data at hospital, state, and national levels is imperative to develop feasible solutions. This analysis could not account for social and community factors that may influence Hispanic patient readmissions, including social determinants of health, provider of implicit/explicit bias, and experiences of discrimination (Velasco-Mondragon et al., 2016). We were unable to account for quality initiatives designed to improve readmissions. Despite the limitations, we are among the first to examine associations between nursing resources and Hispanic surgical patient outcomes using a comprehensive risk adjustment strategy.

Conclusion

Our results showed a significant reduction in the odds of readmission at 7, 30, and 90 days for Hispanic patients relative to their non-Hispanic White counterparts when treated in hospitals with better work environments. Hospitals with good work environments characterized by nurse participation in hospital affairs at all levels, presence of a nursing model of care, and adequate staffing and resources were associated with lower odds of readmission for Hispanic patients. Investments in the work environment may represent a system-based lever to reduce postsurgical readmissions among older Hispanic patients.

Accepted for publication August 6, 2023.

The authors would like to acknowledge Dr. Bridgette M. (Brawner) Rice, PhD, MDiv, The Richard and Marianne Kreider Endowed Professor in Nursing for Vulnerable Populations at the Villanova University Fitzpatrick College of Nursing, and Dr. Karen Hirschman, PhD, MSW, the NewCourtland Term Chair in Health Transitions Research and a Research Professor at the University of Pennsylvania School of Nursing, for their expertise and insights as Dissertation Committee readers. In addition, we want to thank all our Center for Health Outcomes and Policy Research colleagues at the University of Pennsylvania School of Nursing for their continued support.

Research reported in this publication was supported by the following National Institutes of Health (NIH) Grant/Awards: T32 NR007104, R01 NR014855, and R01 AG041099. This research is funded in part by the Rita and Alex Hillman Foundation. The content is solely the responsibility of the authors and does not necessarily represent the official views of the NIH or the Rita and Alex Hillman Foundation.

The RN4CAST-US survey was approved by the University of Pennsylvania Institutional Review Board before it was conducted, and consent was obtained from nurses participating in the survey. Informed consent of patients was waived as all patient data were deidentified and assigned a random ID number for research use. This study was determined exempt by the University of Pennsylvania Institutional Review Board.

The authors have no conflicts of interest to report.

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